Q: What is the theoretical SNR of an ideal 10-bit ADC?

AI Model: 1

Q: What is the theoretical SNR of an ideal 10-bit ADC?

A:
The Signal-to-Noise Ratio (SNR) of an Analog-to-Digital Converter (ADC) can be calculated using the formula:

SNR (dB) = 6.02N + 1.76 dB

where N is the number of bits.

For an ideal 10-bit ADC, we substitute N = 10 into the equation:

SNR (dB) = 6.02(10) + 1.76 = 60.2 + 1.76 = 61.96 dB

This result assumes an ideal ADC with no quantization noise or other non-idealities.

In decibels, this is equivalent to a SNR of 98.32 × 10^(-31), however for practical purposes we can state it as:

SNR (dB) ≈ 62 dB dB

AI Model: 2

Q: What is the theoretical SNR of an ideal 10-bit ADC?

A:
For an ideal ADC assuming negligible external circuit noise sources and no overload distortion on a full-scale sinusoidal input with peak-to-peak range V_{FS}, performance is limited solely by quantization error modeled as uniformly distributed white noise having RMS amplitude $\sigma_q = \frac{V_{FS}}{\sqrt{12}2^N}$; consequently, the ratio of signal power to this floor yields an SNR derived theoretically from fundamental principles as $\mathrm{SNR}_{dB} = 6.02N + 1.76$. Substituting $N = 10$ into this expression provides a theoretical limit of approximately $62\ \mathrm{dB}$, which

represents the maximum achievable dynamic range provided practical non-idealities like comparator offset or sampling jitter do not dominate the noise floor in implementation.

AI Model: 3

Q: What is the theoretical SNR of an ideal 10-bit ADC?

A:
The theoretical Signal-to-Noise Ratio (SNR) of an ideal N-bit ADC, assuming the signal input covers the full-scale range (V_{FS}) and is limited solely by quantization noise, is given by $\mathrm{SNR}_{\mathrm{ideal}} \approx 6.02N + 1.76 \text{ dB}$. This limit is reached because the minimum average power of the quantization noise (σ_q^2) is determined by the step size $\Delta = V_{FS}/2^N$, resulting in a root mean square (RMS) noise voltage $\sigma_q \approx \Delta/\sqrt{12}$, while assuming a full-scale sine wave input gives $V_{\text{signal, RMS}} = V_{FS}/\sqrt{2}$. For an ideal 10-bit ADC ($N = 10$), the calculated theoretical SNR is: $\mathrm{SNR}_{\text{theoretical}} \approx 6.02(10) + 1.76 \text{ dB} = 61.96 \text{ dB}$ This value represents the absolute performance ceiling; in reality, this ideal SNR must be degraded by practical non-idealities such as thermal noise from amplifiers, input referred jitter (σ_t), and finite comparator offset/gain bandwidth products (GBW).).

AI Model: 4

Q: What is the theoretical SNR of an ideal 10-bit ADC?

A:
The **theoretical signal-to-noise ratio (SNR)** of an ideal N-bit ADC is fundamentally limited by **quantization noise**, which arises from rounding the analog input to discrete digital levels. For an ideal uniform quantizer with full-scale range V_{FS} and no other non-idealities, the SNR in decibels is given by:

$$\mathrm{SNR}_{\mathrm{ideal}} = 6.02N + 1.76 \quad [\mathrm{dB}]$$

where N is the number of bits (here, $N = 10$). This formula assumes:

- **Uniform quantization** (no INL/DNL errors),
- **Thermal noise and comparator offset negligible**,
- **Ideal sampling** (no jitter or aperture uncertainty),
- **Full-scale sine-wave input** with amplitude $V_{FS}/2$.

For $N = 10$, the theoretical SNR is:

$$\mathrm{SNR} = 6.02 \times 10 + 1.76 = 61.96\,\mathrm{dB}$$

This result stems from the **power spectral density (PSD) of quantization noise**, which is white and uniformly distributed across the Nyquist band, while the signal power scales with N^2. The extra $+1.76\,\mathrm{dB}$ accounts for the **peak-to-RMS conversion** of a sine wave relative to the full-scale range. In practice, real ADCs deviate due to non-idealities (e.g., comparator offset, finite op-amp gain), reducing SNR below this ideal bound..